Spatial Estimates of SWE for the Rio Grande Basin

April 15, 2025

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Overview Map – Rio Grande Basin Study Area

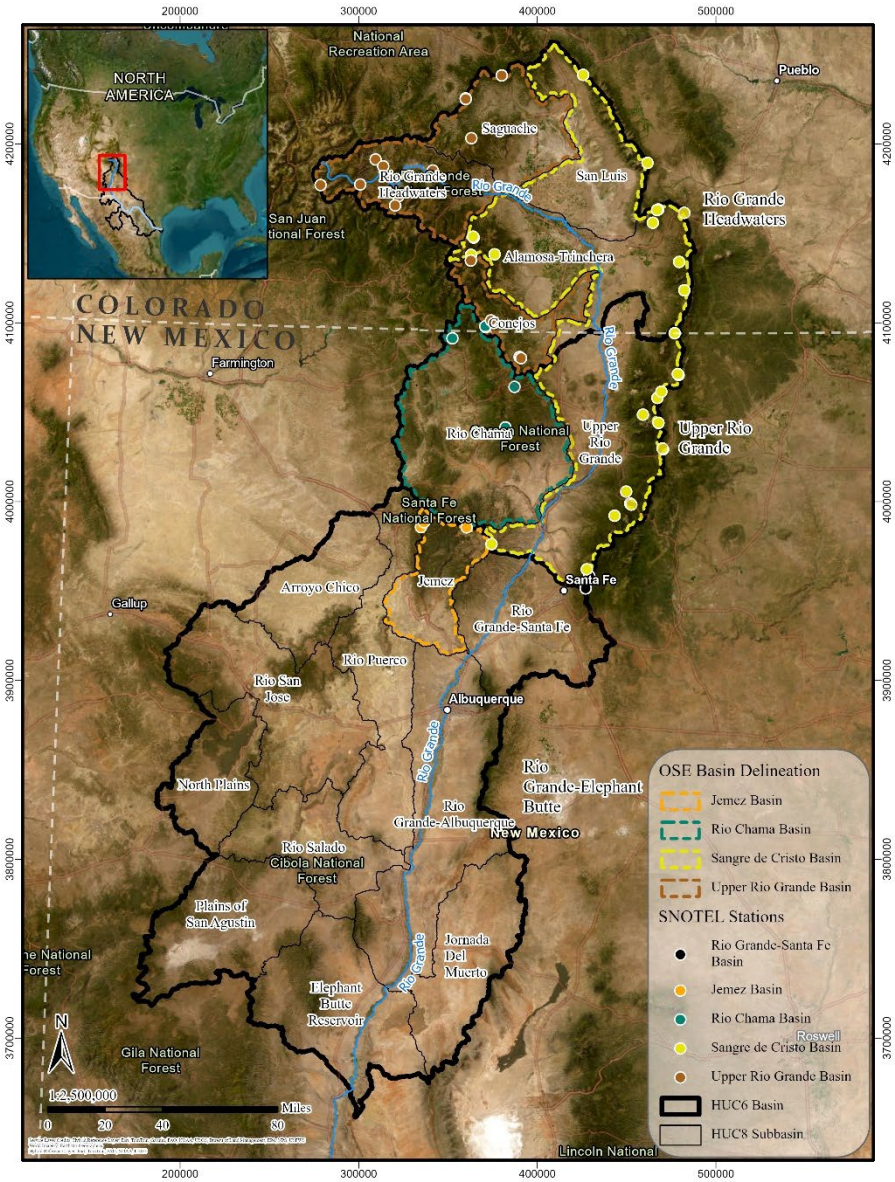

Figure 1. Overview map of the Rio Grande (RG) Basin study area. *The Rio Grande Headwaters, Upper Rio Grande, and Rio Grande-Elephant Butte Basins are presented as the standard HUC6 basins. HUC8 groupings (Jemez – orange, Rio Chama – green, Sangre de Cristo – yellow, and Upper Rio Grande – brown) are symbolized.*

Summary of SNODAS SWE for April 15, 2025

- Spatial estimates of snow water equivalent (SWE) and change in SWE (presented in inches) for the Rio Grande (RG) Basin study area are presented in Figure 2.
- Spring snowmelt persisted throughout the Rio Grande Basin study area due to unusually warm weather, resulting in a decrease in SWE in all three HUC6 Basins since the last report (April 8, 2025).
- **Table 1**: Estimated SWE amount (median and mean in inches (in)), estimated volume (acre-feet (af)), and change in volume (af) since the last report, are shown for the standard Hydrologic Unit Code 6 (HUC6) basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Table 2**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown for the specific HUC8 subbasin groupings (OSE Basin Delineations shown in Figure 1; Upper Rio Grande, Rio Chama, Sangre de Cristo, and Jemez).
- **Table 3**: Estimated SWE amount (median and mean (in)), estimated volume (af), and change in volume (af) since the last report, are shown by basin and elevation band for the HUC6 basins (Rio Grande Headwaters (Figure 3), Upper Rio Grande (Figure 4), and Rio Grande-Elephant Butte (Figure 5)).
- **Rio Grande Headwaters (HUC6)**: SNODAS SWE volume for April 15, 2025, is estimated to be 528,453 af, a decrease of an estimated 343,459 af from April 8, 2025.
- **Upper Rio Grande (HUC6)**: SNODAS SWE volume for April 15, 2025, is estimated to be 133,450 af, a decrease of an estimated 127,076 af from April 8, 2025.
- **Rio Grande-Elephant Butte (HUC6)**: SNODAS SWE volume for April 15, 2025, is estimated to be 131 af, a decrease of an estimated 40,115 af from April 8, 2025.

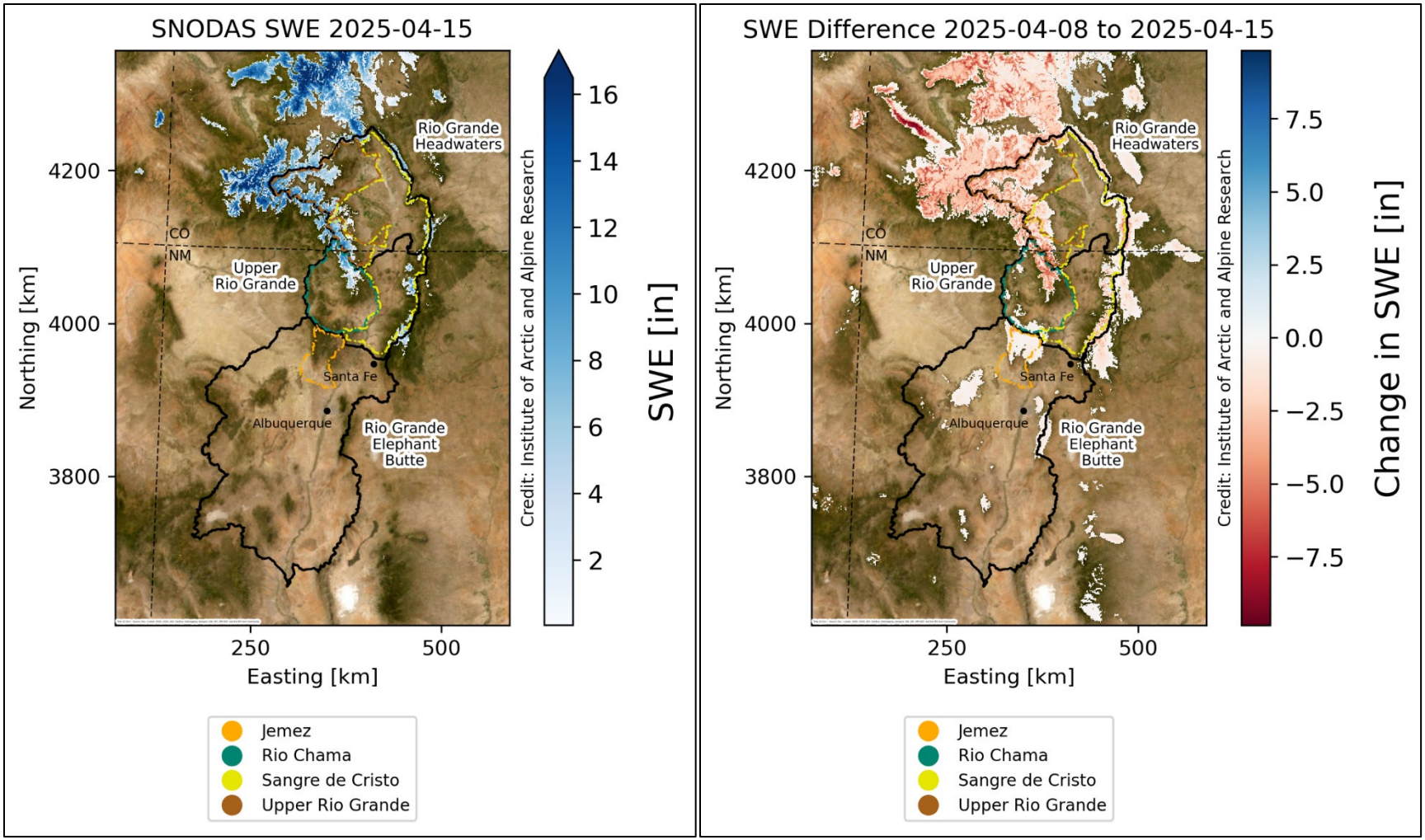

Figure 2. Estimated SNODAS SWE and Change in SNODAS SWE across the Rio Grande Basin study area. *SWE amounts (in) across the RG Basin study area (left); dark blue pixels denote greater SWE amounts. Change in SNODAS SWE amounts (in) across the RG Basin study area from April 8, 2025, to April 15, 2025 (right); red pixels denote a decrease in SWE amounts.*

Introduction

The Rio Grande (RG) is a snowmelt-driven river system with up to 75% of its water originating from seasonal snowpack in the mountains of Colorado and northern New Mexico (Moeser et al., 2021; Figure 2). Snowpack acts as a natural reservoir in the RG Basin, gradually releasing meltwater during the spring and early summer months, sustaining streamflow and providing vital water supply for downstream users. However, regional decreases in snowpack volume and earlier snowmelt timing, especially throughout the western U.S. and New Mexico, can lead to reduced water availability (Mote et al., 2005; EPA, 2016b; Mote et al., 2018; Hale et al., 2023). Long-term changes in snowpack patterns and timing can serve as an indicator of climate change, while understanding short-term changes is critical for monitoring seasonal variability and forecasting seasonal water supply. Monitoring snowpack provides valuable data on shifts in precipitation patterns and temperatures, helping scientists and decision-makers better understand and manage climate-related challenges.

Snow water equivalent (SWE) is the estimate of the depth of liquid water contained in the snowpack. SWE is the standard metric in estimating the volume of meltwater in the snowpack, which plays a critical role in the seasonal snow water supply in the RG Basin. SWE is measured at in-situ Snow Telemetry (SNOTEL) sites throughout the RG Basin and the greater Western U.S. However, the spatial distribution of SWE is complex and often difficult to estimate outside of the in-situ SNOTEL measurements. While SNOTEL provides important point-based measurements, it has been suggested that these data may not be representative of the surrounding snow-covered area (Meromy et al., 2013). Whereas regional tendencies of spatial distribution are consistent, local snowpack volumes are more variable, all of which influences the magnitude and timing of snowmelt-driven runoff.

SWE Analysis Tool

The RG Basin, and more broadly the Western U.S., is experiencing a prolonged and severe drought (EPA, 2016a), exacerbated by a warming climate that increases evapotranspiration, reduces overall snowpack quantity, and reduces the seasonal duration of snowpack. The application of the SWE Analysis Tool (SWEAT) will supply scientists and decision-makers with an improved insight into the spatial distribution of SWE by providing real-time spatial SWE estimates for the RG Basin, its subbasins, and elevation bands within the basin.

Currently, SWEAT incorporates the SNODAS gridded SWE data product (NOHRSC, 2004). Because no spaceborne sensor can directly measure SWE, spatial distribution of SWE must be estimated from remotely sensed observations coupled with modeling approaches. However, gridded SWE products, like SNODAS, are known to have accuracy issues across both mountainous and non-mountainous regions (Mudryk et al., 2025), resulting in a range of spatial SWE estimates for any given location. NM OSE and INSTAAR are working to integrate more data products to build SWEAT based on an ensemble of spatially estimated SWE. By taking an ensemble approach, the goal for SWEAT is to provide spatial SWE estimations for the RG Basin that are, on average, equally or more accurate than any individual model (Thompson, 1977; Arsenault et al., 2015). In addition, the use of a single estimate calculated from an ensemble can reduce confusion for water managers about which product may be performing best, providing a path towards consistency throughout users, and is useful in identifying product outliers.

SNODAS

The SNOw Data Assimilation System (SNODAS) represents a modeling and data assimilation system developed by the National Operational Hydrologic Remote Sensing Center (NOHRSC). SNODAS generates spatial estimates of SWE, snow cover, and snow depth through a combination of remotely sensed snow data and in-situ measurements of SWE from SNOTEL at a 1-km spatial resolution and 24-hour temporal resolution.

The national SNODAS dataset is available through the SNODAS Data Products at National Snow and Ice Data Center (NSIDC), Version 1 (NOHRSC, 2004). SNODAS data can be viewed at the NOHRSC Interactive Snow Information page. Gridded observed snowfall images, seasonal totals, and data downloads in several formats can be found on the National Gridded Snowfall Analysis page.

Refer to the SNODAS User Guide and the NSIDC Special Report 11 (Barrett, 2003) for more information.

Discussion

Figures and tables for the identified basins (Figure 1) are presented below. The median and mean SWE (in) is presented to avoid potentially skewed outliers, most notably due to an uneven distribution of SWE across elevation bands. While the mean value represents the average value, it is highly influenced by outliers and is typically best used for uniform distributions of data. In contrast, the median value represents the middle value and is less sensitive to outliers, which can be more useful in skewed distributions. Note that the basin-wide median and mean values may not be directly comparable with the SNOTEL basin-wide values. If comparisons are made with SNOTEL sites, a better comparison may be made across elevation bands (Table 3).

Figure 3 (Rio Grande Headwaters), Figure 4 (Upper Rio Grande), and Figure 5 (Rio Grande-Elephant Butte) show the spatial distribution of SWE amount (in), change in SWE volume (af) since the last report, and SWE volume (af) across subbasin and elevation. Since the last report, spring snowmelt persisted throughout the Rio Grande Basin study area due to unusually warm weather, resulting in a decrease in SWE in all three HUC6 Basins.

Estimated SWE amounts (in), volumes (af), and change in volumes (af) since the last report are presented in the tables below.

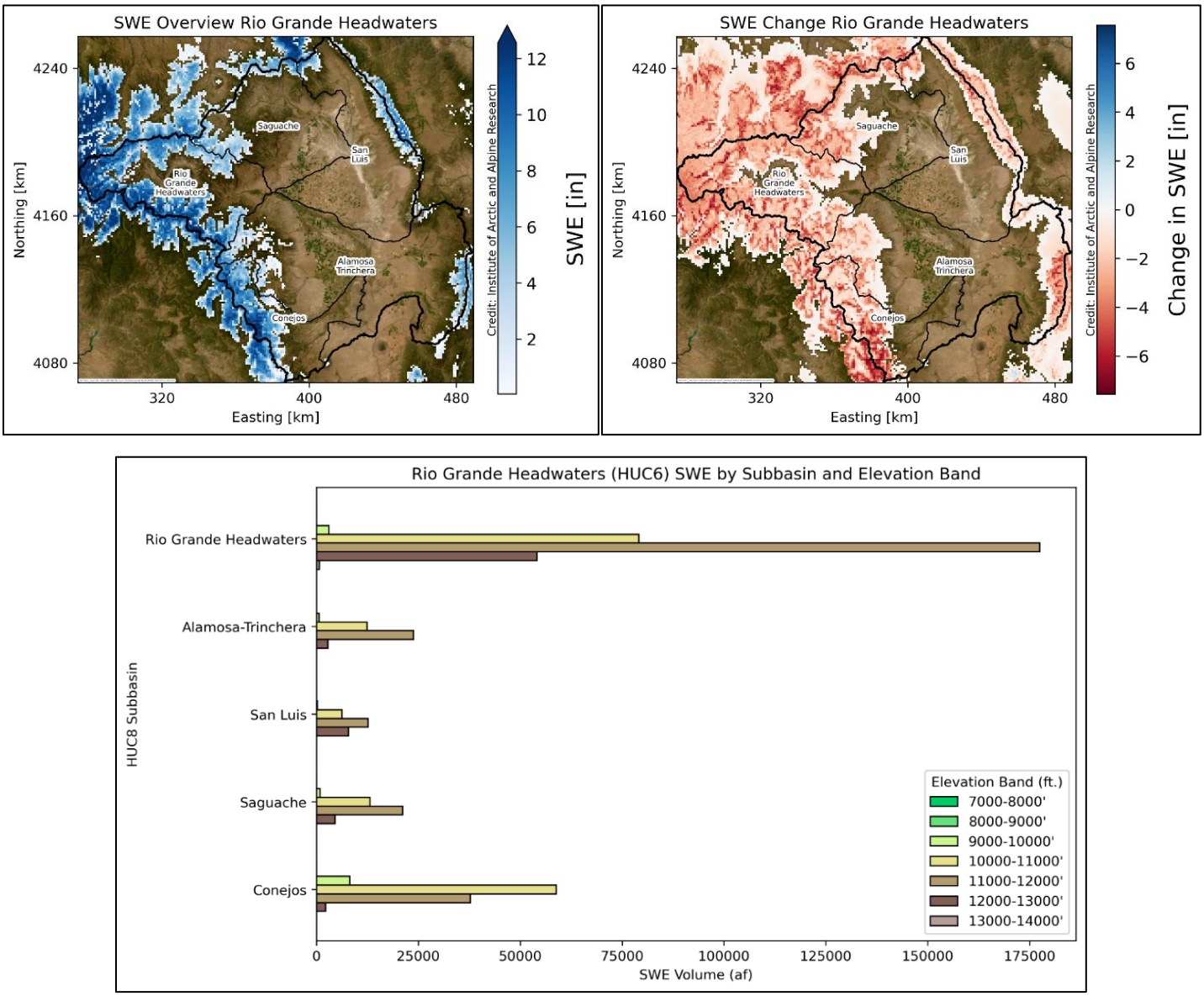

Figure 1. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande Headwaters Basin for April 15, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Rio Grande Headwaters Basin (HUC6), including the Rio Grande Headwaters, Alamosa-Trinchera, San Luis, Saguache, and Conejos subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande Headwaters Basin (bottom).*

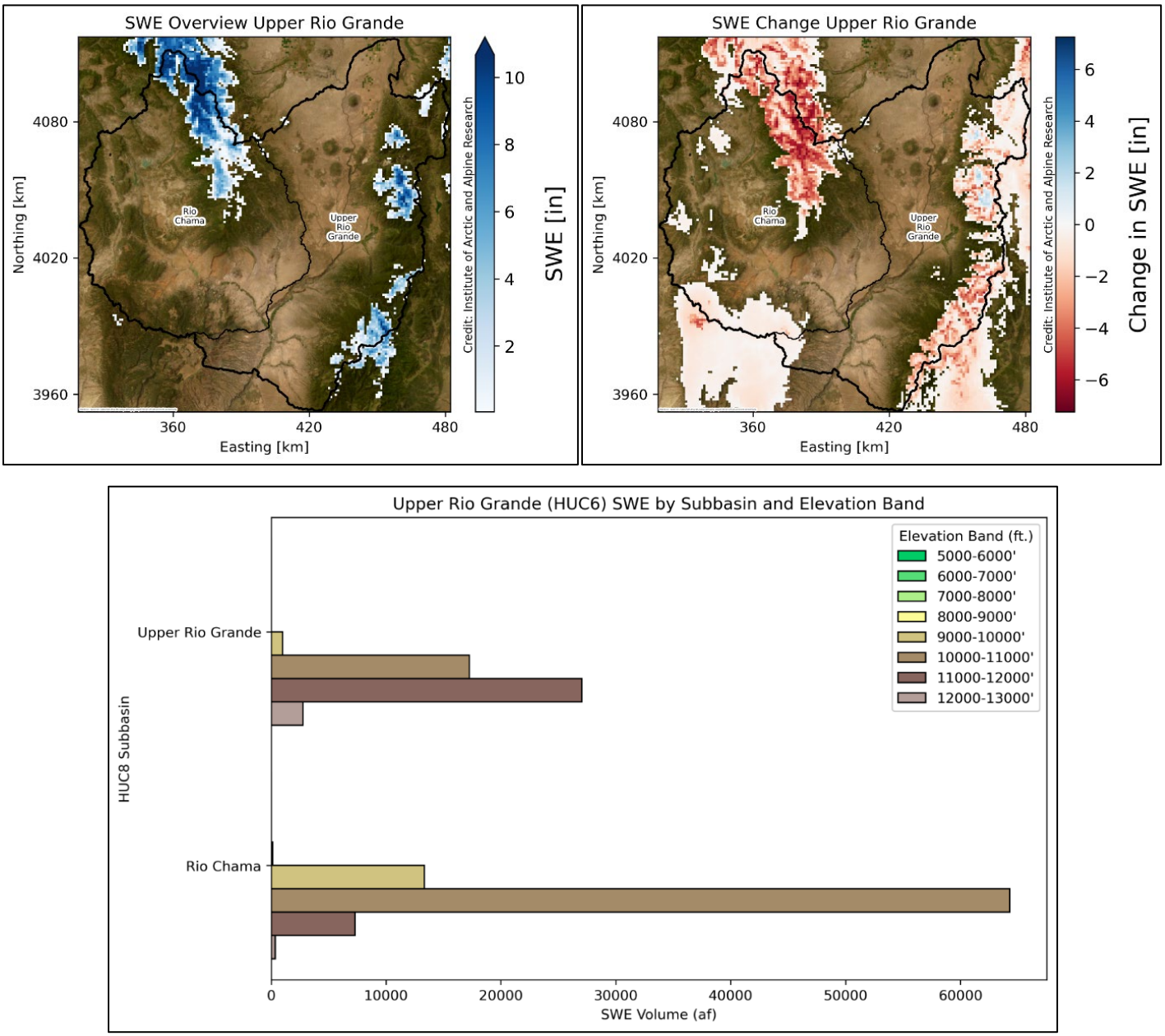

Figure 2. Estimated SNODAS SWE (in) and SWE volume (af) across the Upper Rio Grande Basin for April 15, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across the Upper Rio Grande Basin (HUC6), including the Upper Rio Grande and Rio Chama subbasins. SWE volume (af) by subbasin and elevation band across the Upper Rio Grande Basin (bottom).*

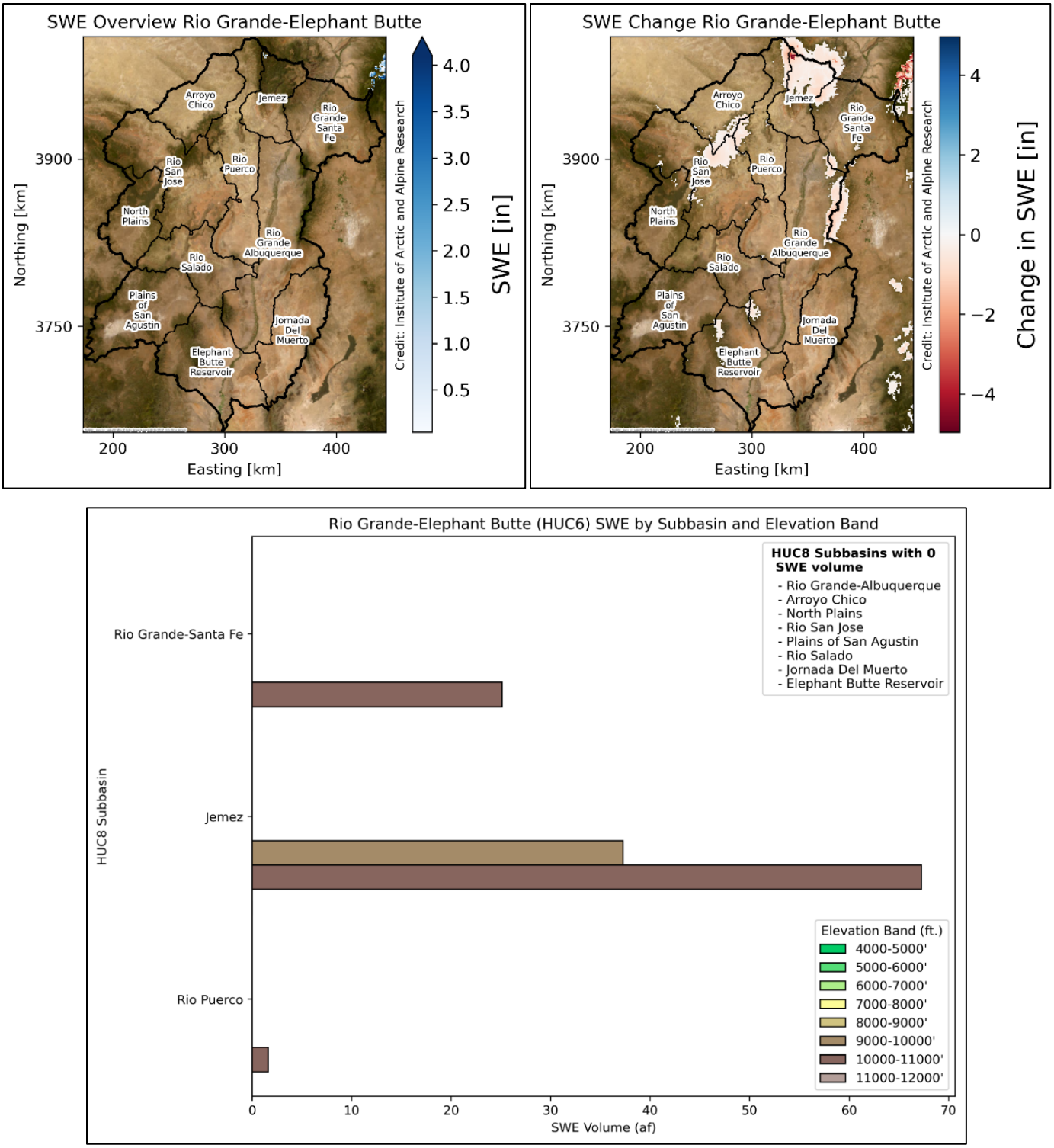

Figure 3. Estimated SNODAS SWE (in) and SWE volume (af) across the Rio Grande-Elephant Butte Basin for April 15, 2025. *SWE amount (top left) and change in SWE since the last report (top right) across Rio Grande-Elephant Butte (HUC6), including the Rio Grande-Santa Fe, Jemez, Rio Grande-Albuquerque, Arroyo Chico, North Plains, Rio San Jose, Plains of San Agustin, Rio Salado, Jornada Del Muerto, Elephant Butte Reservoir, and Rio Puerco subbasins. SWE volume (af) by subbasin and elevation band across the Rio Grande-Elephant Butte Basin (bottom).*

Tables

Table 1. Estimated SWE by HUC6 Basin for April 15, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each HUC8 basin (bold).*

HUC6 Basin	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 8, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	3.1	4.3	314,585	-145,676
	Alamosa-Trinchera	0.0	0.3	39,771	-56,637
	San Luis	0.0	0.3	27,133	-26,205
	Saguache	0.0	0.6	39,835	-60,879
	Conejos	0.0	2.6	107,129	-54,062
	Total	**0.0**	**1.3**	**528,453**	**-343,459**
Upper Rio Grande	Upper Rio Grande	0.0	0.3	48,057	-47,846
	Rio Chama	0.0	0.5	85,393	-79,230
	Total	**0.0**	**0.4**	**133,450**	**-127,076**
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	0.0	0.0	25	-4,097
	Jemez	0.0	0.0	105	-13,978
	Rio Grande-Albuquerque	0.0	0.0	0	-7,187
	Arroyo Chico	0.0	0.0	0	-2,152
	North Plains	0.0	0.0	0	-142
	Rio San Jose	0.0	0.0	0	-5,499
	Plains of San Agustin	0.0	0.0	0	-24
	Rio Salado	0.0	0.0	0	-218
	Jornada Del Muerto	0.0	0.0	0	0
	Elephant Butte Reservoir	0.0	0.0	0	-931
	Rio Puerco	0.0	0.0	2	-5,886
	Total	**0.0**	**0.0**	**131**	**-40,115**

Table 2. Estimated SWE by Specified Subbasin Groupings for April 15, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by HUC8 subbasin and totaled for each specified subbasin grouping (bold).*

Subbasin Grouping	HUC8 Subbasin	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 8, 2025 (af)
Upper Rio Grande	Rio Grande Headwaters	3.1	4.3	314,585	-145,676
	Saguache	0.0	0.6	39,835	-60,879
	Conejos	0.0	2.6	107,129	-54,062
	Total	**0.0**	**2.5**	**461,549**	**-258,790**
Sangre de Cristo	San Luis	0.0	0.3	27,133	-26,205
	Alamosa-Trinchera	0.0	0.3	39,771	-56,637
	Upper Rio Grande	0.0	0.3	48,057	-47,846
	Total	**0.0**	**0.3**	**114,961**	**-129,614**
Rio Chama		**0.0**	**0.5**	**85,393**	**-80,350**
Jemez		**0.0**	**0.0**	**105**	**-14,233**

Table 3. Estimated SWE by HUC6 Basin and Elevation Band for April 15, 2025. *Shown are median SWE (in.), mean SWE (in.), estimated volume (af), and volume change (af) since the last report. Values are separated by elevation band (1000 ft. increments) and HUC8 subbasin.*

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 8, 2025 (af)
Rio Grande Headwaters	Rio Grande Headwaters	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-441
		9000-10000'	0.0	0.2	3,064	-12,285
		10000-11000'	2.7	3.6	79,142	-65,283
		11000-12000'	8.5	8.1	177,488	-51,661
		12000-13000'	9.7	9.0	54,155	-15,720
		13000-14000'	8.3	6.0	736	-285
	Alamosa-Trinchera	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-537
		9000-10000'	0.0	0.0	627	-10,172
		10000-11000'	0.0	1.0	12,455	-25,723
		11000-12000'	2.2	2.8	23,832	-17,249
		12000-13000'	0.3	1.5	2,857	-2,859
		13000-14000'	0.0	0.0	0	-97
	San Luis	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-16
		9000-10000'	0.0	0.0	214	-2,325
		10000-11000'	0.0	1.0	6,297	-10,894
		11000-12000'	1.9	2.7	12,689	-9,328
		12000-13000'	1.8	3.0	7,897	-3,600
		13000-14000'	0.0	0.6	36	-42
	Saguache	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-41
		9000-10000'	0.0	0.1	912	-12,013
		10000-11000'	0.0	1.1	13,177	-34,607
		11000-12000'	4.7	4.6	21,201	-12,171
		12000-13000'	4.8	4.9	4,542	-2,004
		13000-14000'	0.1	0.1	2	-42
	Conejos	7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	36	-204
		9000-10000'	0.0	0.8	8,152	-19,070
		10000-11000'	6.6	6.2	58,870	-28,030

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 8, 2025 (af)
	Conejos	11000-12000'	8.3	7.7	37,776	-6,320
		12000-13000'	6.0	5.9	2,296	-437
Upper Rio Grande	Upper Rio Grande	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-477
		9000-10000'	0.0	0.0	996	-12,915
		10000-11000'	0.0	1.0	17,252	-28,108
		11000-12000'	3.0	3.4	27,040	-6,007
		12000-13000'	1.7	2.4	2,769	-340
	Rio Chama	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	-231
		8000-9000'	0.0	0.0	112	-3,516
		9000-10000'	0.0	0.6	13,341	-40,508
		10000-11000'	5.8	5.7	64,295	-33,897
		11000-12000'	9.2	9.1	7,288	-1,055
		12000-13000'	8.7	8.7	357	-23
Rio Grande-Elephant Butte	Rio Grande-Santa Fe	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	-9
		7000-8000'	0.0	0.0	0	-1,132
		8000-9000'	0.0	0.0	0	-2,035
		9000-10000'	0.0	0.0	0	-511
		10000-11000'	0.0	0.1	25	-349
		11000-12000'	0.0	0.0	0	-62
	Jemez	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	-1
		7000-8000'	0.0	0.0	0	-1,116
		8000-9000'	0.0	0.0	0	-6,831
		9000-10000'	0.0	0.0	37	-4,526
		10000-11000'	0.0	0.1	67	-1,504
	Rio Grande-Albuquerque	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	-251
		7000-8000'	0.0	0.0	0	-4,141

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 8, 2025 (af)
Rio Grande-Elephant Butte	Rio Grande-Albuquerque	8000-9000'	0.0	0.0	0	-2,057
		9000-10000'	0.0	0.0	0	-676
		10000-11000'	0.0	0.0	0	-62
	Arroyo Chico	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	-487
		8000-9000'	0.0	0.0	0	-1,465
		9000-10000'	0.0	0.0	0	-184
		10000-11000'	0.0	0.0	0	-16
	North Plains	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	-95
		8000-9000'	0.0	0.0	0	-47
	Rio San Jose	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	-1
		7000-8000'	0.0	0.0	0	-1,108
		8000-9000'	0.0	0.0	0	-3,107
		9000-10000'	0.0	0.0	0	-998
		10000-11000'	0.0	0.0	0	-285
	Plains of San Agustin	6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-24
		9000-10000'	0.0	0.0	0	0
	Rio Salado	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	-32
		8000-9000'	0.0	0.0	0	-137
		9000-10000'	0.0	0.0	0	-49
	Jornada Del Muerto	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	0
	Elephant Butte Reservoir	4000-5000'	0.0	0.0	0	0

HUC6 Basin	HUC8 Subbasin	Elevation Band (ft.)	Median SWE (in.)	Mean SWE (in.)	Estimated Volume (af)	Volume Change since April 8, 2025 (af)
Rio Grande-Elephant Butte	Elephant Butte Reservoir	5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	0
		7000-8000'	0.0	0.0	0	0
		8000-9000'	0.0	0.0	0	-433
		9000-10000'	0.0	0.0	0	-486
		10000-11000'	0.0	0.0	0	-13
	Rio Puerco	4000-5000'	0.0	0.0	0	0
		5000-6000'	0.0	0.0	0	0
		6000-7000'	0.0	0.0	0	-3
		7000-8000'	0.0	0.0	0	-1,097
		8000-9000'	0.0	0.0	0	0
		9000-10000'	0.0	0.0	0	0
		10000-11000'	0.0	0.0	0	0

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References

Arsenault, R., Gatien, P., Renaud, B., Brissette, F., & Martel, J. L. (2015). A comparative analysis of 9 multi-model averaging approaches in hydrological continuous streamflow simulation. *Journal of Hydrology*, 529, 754-767.

Barrett, Andrew. (2003). *National Operational Hydrologic Remote Sensing Center Snow Data Assimilation System (SNODAS) Products at NSIDC*. NSIDC Special Report 11. Boulder, CO USA: National Snow and Ice Data Center. 19 pp.

Environmental Protection Agency [EPA]. (2016a). *Climate Change Indicators: Drought*. Web update: June 2024. Available at: https://www.epa.gov/climate-indicators/climate-change-indicators-drought. Accessed: March 31, 2025.

Environmental Protection Agency [EPA]. (2016b) *Climate Change Indicators: Snowpack*. Web update: June 2024. Available at: https://www.epa.gov/climate-indicators/climate-change-indicators-snowpack. Accessed: March 31, 2025.

Hale, K.E., Jennings, K.S., Musselman, K.N., Livneh, B., Molotch, N.P. (2023). Recent decreases in snow water storage in western North America. *Communications Earth & Environment*, 4(1), p. 170. Available at: https://doi.org/10.1038/s43247-023-00751-3.

Meromy, L., Molotch, N.P., Link, T.E., Fassnacht, S.R., & Rice, R. (2013). Subgrid variability of snow water equivalent at operational snow stations in the western USA. *Hydrological Processes, 27*, 2383-2400, doi: 10.1002/hyp.9355.

Moeser, C. David., Chavarria, S. B., & Wootten, A. M. (2021). *Streamflow response to potential changes in climate in the Upper Rio Grande Basin* (USGS Numbered Series No. 2021–5138; Scientific Investigations Report). U.S. Geological Survey. https://www.usgs.gov/publications/streamflow-response-potential-changes-climate-upper-rio-grande-basin.

Mote, P. W., Hamlet, A. F., Clark, M. P., & Lettenmaier, D. P. (2005). Declining mountain snowpack in western North America. *Bulletin of the American Meteorological Society, 86*(1), 39–50. https://doi.org/10.1175/BAMS-86-1-39.

Mote, P. W., Li, S., Lettenmaier, D. P., Xiao, M., & Engel, R. (2018). Dramatic declines in snowpack in the western US. *Climate and Atmospheric Science*, 1, 2. https://doi.org/10.1038/s41612-018-0012-1.

Mudryk, L., Mortimer, C., Derksen, C., Elias Chereque, A., Kushner, P. (2025). Benchmarking of snow water equivalent (SWE) products based on outcomes of the SnowPEx+ Intercomparison Project. *The Cryosphere, 19*(1), pp. 201–218. Available at: https://doi.org/10.5194/tc-19-201-2025.

NASA Wester Water Applications Office [WWAO] and Daniel B. Stephens & Associates, Inc. [DBS&A]. (2022). *Rio Grande River Basin Needs Assessment Report.* NASA WWAO Assessment Report. Available at: https://wwao.jpl.nasa.gov/documents/138/Water-Needs-Rio-Grande-Report-2022.pdf.

National Operational Hydrologic Remote Sensing Center [NOHRSC]. (2004). *Snow Data Assimilation System (SNODAS) Data Products at NSIDC, Version 1*. SWE Dataset. Boulder, Colorado USA. NSIDC: National Snow and Ice Data Center. https://doi.org/10.7265/N5TB14TC. Accessed: April 1, 2025, and as needed.

Thompson, P.D. (1977). How to improve accuracy by combining independent forecasts. *Monthly Weather Review, 105*(2), pp.228-229.